TROPIC Clinical Analysis Report

Study EFC6193 / XRP6258 -- Abbreviated Clinical Study Summary

Study Title: Prednisone plus Cabazitaxel or Mitoxantrone for mCRPC Progressing After Docetaxel

Sponsor: Sanofi-Aventis | Phase: III, Open-Label RCT | NCT: 00417079

Publication: de Bono JS et al., Lancet 2010;376(9747):1147-1154

***Mixed real + synthetic data.** All MP-arm statistics (N=371) are derived from the official Sanofi de-identified SDTM public data release (2013) -- real patient-level data. The Cabazitaxel (CbzP, N=378) arm is a **synthetic, illustrative** cohort generated by proportional-hazards time-scaling of the real MP arm (real MP event times ÷ published HR, censoring calibrated to published event counts) plus fixed-seed sampling from published Lancet 2010 Table 1/Table 2 marginal distributions. It is **not real data and not the Guyot et al. (2012) KM-digitisation method.** Any CbzP-vs-MP comparison below is **circular by construction** and is shown only to demonstrate the analysis pipeline -- **not** as a clinical finding.*

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1. Study Population

The Safety Population consisted of 371 patients in the Mitoxantrone + Prednisone (MP) arm, all with metastatic castration-resistant prostate cancer (mCRPC) progressing during or after docetaxel-based chemotherapy.

Characteristic	MP Arm (N=371)
ECOG Performance Status ?1	Majority
Prior docetaxel progression	100% (eligibility criterion)
Measurable disease	Subset (MEASDISF = Y)
Visceral disease	Subset (VISCFL = Y)
Baseline PSA (median)	~110 ng/mL
Baseline ALP (median)	~140 U/L

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2. Overall Survival -- Pipeline Demonstration *(synthetic comparator)*

The OS analysis exercises the stratified Cox / log-rank machinery and the hierarchical step-down gatekeeping logic (ICH E9). The CbzP column below is computed from synthetic data and is not a measure of treatment effect.

Statistic	Synthetic CbzP (N=378)[dagger]	Real MP (N=371)	Published (de Bono 2010)
Median OS	21.7 months (synthetic)	12.7 months (real, 95% CI: 11.8-14.1)	15.1 mo vs 12.7 mo

Statistic	Synthetic CbzP (N=378)[dagger]	Real MP (N=371)	Published (de Bono 2010)
OS Events / N	200/378 (synthetic, calibrated)	266/371 (real)	200/378 vs 266/371
Pipeline HR (synthetic)[double-dagger]	0.43 (95% CI: 0.35-0.52)	Reference	published HR 0.70 (0.59-0.83)
Log-Rank p-value[double-dagger]	<0.0001	--	0.0004

***This is not a finding.** Because the synthetic CbzP times are the real MP times divided by an assumed HR, a "significant" separation is guaranteed by construction and is non-inferential. Note also that the procedure **does not reproduce** the published cabazitaxel OS (synthetic 21.7 mo and HR 0.43 vs published 15.1 mo and HR 0.70); the synthetic arm overshoots the real drug's effect and must not be cited as a re-analysis result.*

[dagger]Synthetic, illustrative -- not real patient data. [double-dagger]Circular by construction; descriptive of synthetic data only.

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3. Secondary Endpoints -- Pipeline Demonstration *(synthetic comparator)*

The gatekeeping logic is exercised on these endpoints; the CbzP column is synthetic and the HRs/p-values are circular by construction (non-inferential). Values are the live output of 09_tfl/tfl_generation.R (see 09_tfl/output/tables/T-11-Efficacy_Tables.txt).

Endpoint	Synthetic CbzP[dagger]	Real MP	Pipeline HR (95% CI)[double-dagger]	Pipeline p[double-dagger]	Published (de Bono 2010)
Progression-Free Survival	1.9 mo (synthetic)	1.4 mo (real)	0.66 (0.56-0.78)	<0.0001	median 2.8 mo, HR 0.74
Time to PSA Progression	2.8 mo (synthetic)	2.2 mo (real, 95% CI 1.7-3.3)	0.84 (0.71-0.99)	0.0362	median 6.4 mo, HR 0.75
Time to Tumour Progression	3.8 mo (synthetic)§	2.3 mo (real)§	0.67 (0.54-0.83)	0.0003	median 8.8 mo, HR 0.61
PSA Response (?50% decrease)	39.2% (148/378, synthetic)	18.6% (69/371, real)	Fisher's exact	<0.0001	39% vs 24% (p = 0.0002)
Overall Response Rate (ORR, confirmed)	12.8% (23/179, synthetic)§	7.9% (16/203, real)§	Fisher's exact	0.1296	14.4% vs 4.4% (p = 0.005)

[dagger]Synthetic, illustrative -- not real patient data. [double-dagger]Circular by construction; descriptive of synthetic

data only, not a treatment-effect estimate. §Restricted to Measurable Disease Subpopulation (N=179 synthetic CbzP, N=203 real MP). The synthetic CbzP medians do not match the published cabazitaxel medians (the reconstruction overshoots; see §8).

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4. Safety -- Adverse Events (Safety Population)

Treatment-emergent adverse events (TEAEs) were defined as events occurring on or after the first dose date (TRTEMFL = "Y" in ADAE).

4.1 Overall TEAE Incidence

Category	**CbzP (N=378)**	**%**	**MP (N=371)**	**%**
Any TEAE	364	**96%**	328	**88%**
Any Grade ?3 TEAE	310	**82%**	147	**40%**
Any Serious TEAE (SAE)	145	**38%**	78	**21%**
Any TEAE leading to discontinuation	68	**18%**	0	**0%**

4.2 Grade ?3 TEAEs by System Organ Class (Top 6)

System Organ Class	**CbzP (n, %)**	**MP (n, %)**
Blood & Lymphatic System Disorders	293 (78%)	39 (11%)
Gastrointestinal Disorders	34 (9%)	6 (2%)
General Disorders & Admin Site Conditions	35 (9%)	36 (10%)
Musculoskeletal & Connective Tissue Disorders	18 (5%)	35 (9%)
Infections & Infestations	8 (2%)	19 (5%)
Nervous System Disorders	4 (1%)	14 (4%)

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5. Laboratory Toxicity -- CTCAE Grade Shift

Baseline to worst post-baseline CTCAE grade shifts were derived from the ADLB dataset using BASEFL, ANL01FL, and ATOXGR variables.

5.1 ANC / Neutrophils -- Key Finding

Of 371 safety-evaluable patients in the CbzP arm, 321 (86.5%) experienced Grade 3/4 neutropenia compared to 154 (41.5%) in the MP arm, highlighting the hematological toxicity signature of Cabazitaxel.

5.2 Haemoglobin

Grade 3/4 anemia occurred in 34 (9.2%) patients in the CbzP arm compared to 9 (2.4%) in the MP arm.

5.3 Platelets

Thrombocytopenia was rare, with Grade 3/4 thrombocytopenia occurring in 16 (4.3%) CbzP patients vs 5 (1.3%) MP patients.

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6. Exposure Analysis (ADEX)

Cycle-by-cycle exposure was captured in ADEX across parameters including:

* **RDI** (Relative Dose Intensity) -- ratio of actual to planned cumulative dose
* **NCYCLE** -- number of treatment cycles completed per patient
* **CUMDOSE** -- cumulative dose received
* **NDELDOSE / NREDDOSE** -- cycles with delays or dose reductions

FDA Project Optimus alignment: RDI in Cycle 1 was used as the E-R exposure proxy in the scatter plot analysis (F-17-1).

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7. Pipeline Technical Summary

Layer	**Technology**	**Standard**
Source Data	SAS7BDAT (Sanofi / Project Data Sphere)	CDISC SDTMIG v3.1.1 (trial-era)
ADaM Production	SAS 9.4	ADaMIG v1.3
Independent Validation	R 4.6.0 / Pharmaverse	ADaMIG v1.3
Reconciliation	`diffdf` package	100% cell-by-cell match
TFL Generation	ggplot2, survival, patchwork	ICH E3 / NEJM style
Orchestration	Python 3.10 (cibuild.py)	12-stage CI pipeline

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8. Data Provenance & Limitations

***Real data (MP arm):** All 371 MP-arm patients, 5,428 AE records, 266 OS events, and ~79,000 laboratory records are derived directly from the official Sanofi de-identified public SDTM release (dated June 2013).*

***Synthetic comparator (CbzP arm):** The Cabazitaxel arm was not included in the Sanofi public data release. The CbzP arm used in figures and comparative tables is **synthetic**, generated at the ADaM layer by **proportional-hazards time-scaling of the real MP arm** (real MP event times divided by the published hazard ratio, with censoring calibrated to published event counts) and fixed-seed sampling from published Lancet 2010 Table 1/Table 2 marginal distributions. **This is not the Guyot et al. (2012) KM-digitisation algorithm**, and the synthetic arm does **not** reproduce the published cabazitaxel efficacy (it overshoots -- synthetic OS median 21.7 mo / HR 0.43 vs published 15.1 mo / HR 0.70). All CbzP-vs-MP comparisons are circular by construction and are illustrative only.*

***Single source of truth.** Every count, percentage, median, HR and p-value in this report is produced by `09_tfl/tfl_generation.R` and written to `09_tfl/output/tables/*.txt`. Narrative numbers are transcribed from those files; the generated tables govern in case of any discrepancy.*

Reference

de Bono JS, Oudard S, Ozguroglu M, et al. Prednisone plus cabazitaxel or mitoxantrone for metastatic castration-resistant prostate cancer progressing after docetaxel treatment: a randomised open-label trial. Lancet. 2010;376(9747):1147-1154.

* Local Copy: [de_bono_lancet_2010.pdf](file:///Users/apple/Desktop/TROPIC/01_raw_source/reference_literature/de_bono_lancet_2010.pdf)